

Single-Area OSPF IPv4

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Purpose

Setting up a single-area OSPF for multiple networks allows me to review my networking knowledge and deepen my understanding of single-area OSPF. This lab involves the setting up a network, diagnosing and correcting any issues that may occur when building this network, and teaches me to professionally present my process and abilities to configure a network.

Background Information

The primary concept of this lab is OSPF or Open Shortest Path First, first created by the Internet Engineering Task Force (ISTF) who recognized that RIP on its own would not be able to keep up with and maintain the systems on the internet. OSPF was designed to allow routers to dynamically figure out what the shortest path is between any two networks, hence the name. A few key characteristics or advantages of OSPF is that it has unlimited hop counts, both IPv4 and IPv6 routing protocols, can run on most routers, and can provide route summarization which saves memory, bandwidth (a range of frequencies on a certain band), and CPU cycles (less packets are processed and routing tables are smaller) on the router, providing stability on networks.

Lab Summary

In this lab, I created seven networks to connect four Cisco 4321 routers with a NIM-2T added to them and four computers so they could communicate with each other quickly and efficiently in Packet Tracer. Four of the networks connected each PC to a different router through the router interface G0/0/0 and the other three networks connected the four routers to each other through the serial interfaces S0/1/0 and S0/1/1. For all the networks, the base of their IP addresses was 192.168 and the third number was different for each router, going from 1-7. The last number was chosen from 1-3, with 3 being reserved for the computers and 1-2 being for the interfaces on the routers in the same network.

Lab Commands

Clock Rate: Determines the number of bits that can be delivered over a certain amount of time and synchronize the two ends of the serial link on the routers

Network *IP Address Wildcard Mask Area*: Identify which network and into what OSPF area the interface will go into

Router-ID: Statically sets up the router's ID which can be used to identify the network device

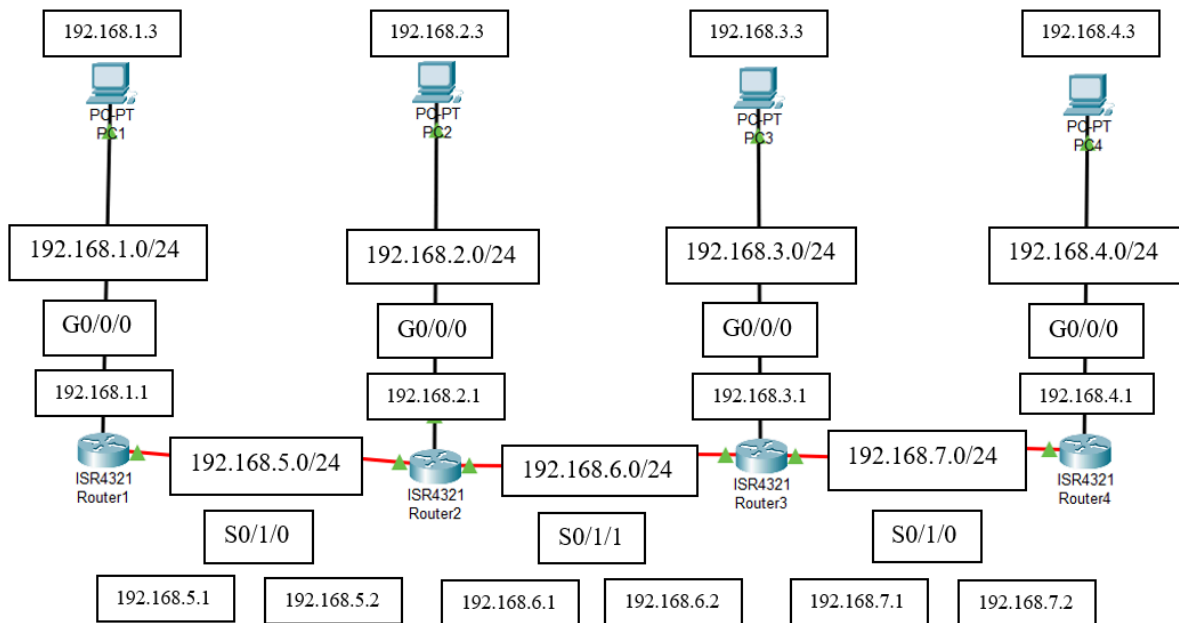
Router OSPF: Enables or starts the OSPF routing protocol on the router

Show IP OSPF Interface: Shows information of all the interfaces on the router that have OSPF enabled

Show IP OSPF Neighbor: Displays the information of surrounding networks (aka neighbors) that this network is next to

Show IP Route: Presents the routing table which has all the networks that the router can reach

Network Diagram



Configurations

- **Router 1**

Show Run

```
interface GigabitEthernet0/0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto

interface Serial0/1/0
 ip address 192.168.5.1 255.255.255.0

router ospf 10
 router-id 1.1.1.1
 log-adjacency-changes
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.5.0 0.0.0.255 area 0
```

Show IP OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
 Internet address is 192.168.1.1/24, Area 0
```

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```

    Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST,
Cost: 1
    Transmit Delay is 1 sec, State DR, Priority 1
    Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
    No backup designated router on this network
    Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:03
    Index 1/1, flood queue length 0
    Next 0x0(0)/0x0(0)
    Last flood scan length is 1, maximum is 1
    Last flood scan time is 0 msec, maximum is 0 msec
    Neighbor Count is 0, Adjacent neighbor count is 0
    Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
    Internet address is 192.168.5.1/24, Area 0
    Process ID 10, Router ID 1.1.1.1, Network Type POINT-TO-POINT,
Cost: 64
    Transmit Delay is 1 sec, State POINT-TO-POINT,
    Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:03
    Index 2/2, flood queue length 0
    Next 0x0(0)/0x0(0)
    Last flood scan length is 1, maximum is 1
    Last flood scan time is 0 msec, maximum is 0 msec
    Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
    Suppress hello for 0 neighbor(s)

```

Show IP OSPF Neighbor

Neighbor ID Interface	Pri	State	Dead Time	Address
2.2.2.2 Serial0/1/0	0	FULL/ -	00:00:35	192.168.5.2

Show IP Route

Gateway of last resort is not set

```

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected,
GigabitEthernet0/0/0

```

```

O    192.168.2.0/24 [110/65] via 192.168.5.2, 01:11:08,
Serial0/1/0
O    192.168.3.0/24 [110/129] via 192.168.5.2, 01:10:58,
Serial0/1/0
O    192.168.4.0/24 [110/193] via 192.168.5.2, 01:10:58,
Serial0/1/0
    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/24 is directly connected, Serial0/1/0
L    192.168.5.1/32 is directly connected, Serial0/1/0
O    192.168.6.0/24 [110/128] via 192.168.5.2, 01:11:08,
Serial0/1/0
O    192.168.7.0/24 [110/192] via 192.168.5.2, 01:10:58,
Serial0/1/0

```

- Router 2

Show Run

```

interface GigabitEthernet0/0/0
 ip address 192.168.2.1 255.255.255.0
 duplex auto
 speed auto

interface Serial0/1/0
 ip address 192.168.5.2 255.255.255.0
 clock rate 56000

interface Serial0/1/1
 ip address 192.168.6.1 255.255.255.0

router ospf 20
 router-id 2.2.2.2
 log-adjacency-changes
 network 192.168.5.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
 network 192.168.6.0 0.0.0.255 area 0

```

Show IP OSPF Interface

```

GigabitEthernet0/0/0 is up, line protocol is up
 Internet address is 192.168.2.1/24, Area 0
 Process ID 20, Router ID 2.2.2.2, Network Type BROADCAST,
 Cost: 1
 Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 2.2.2.2, Interface address 192.168.2.1
 No backup designated router on this network

```

```

Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Internet address is 192.168.6.1/24, Area 0
  Process ID 20, Router ID 2.2.2.2, Network Type POINT-TO-POINT,
Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Hello due in 00:00:05
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
  Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Internet address is 192.168.5.2/24, Area 0
  Process ID 20, Router ID 2.2.2.2, Network Type POINT-TO-POINT,
Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Hello due in 00:00:04
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
  Adjacent with neighbor 1.1.1.1
  Suppress hello for 0 neighbor(s)

```

Show Ip OSPF Neighbor

Neighbor ID	Pri	State	Dead Time	Address
Interface				
1.1.1.1	0	FULL/ -	00:00:38	192.168.5.1
Serial0/1/0				


```
3.3.3.3          0    FULL/  -          00:00:36    192.168.6.2
Serial0/1/1
```

Show IP Route

Gateway of last resort is not set

```
O    192.168.1.0/24 [110/65] via 192.168.5.1, 00:08:17,
Serial0/1/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected,
GigabitEthernet0/0/0
L    192.168.2.1/32 is directly connected,
GigabitEthernet0/0/0
O    192.168.3.0/24 [110/65] via 192.168.6.2, 00:08:17,
Serial0/1/1
O    192.168.4.0/24 [110/129] via 192.168.6.2, 00:08:17,
Serial0/1/1
    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/24 is directly connected, Serial0/1/0
L    192.168.5.2/32 is directly connected, Serial0/1/0
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/24 is directly connected, Serial0/1/1
L    192.168.6.1/32 is directly connected, Serial0/1/1
```

• Router 3

Show Run

```
interface GigabitEthernet0/0/0
 ip address 192.168.3.1 255.255.255.0
 duplex auto
 speed auto

interface Serial0/1/0
 ip address 192.168.7.1 255.255.255.0

interface Serial0/1/1
 ip address 192.168.6.2 255.255.255.0
 clock rate 56000

router ospf 30
 router-id 3.3.3.3
 log-adjacency-changes
 network 192.168.6.0 0.0.0.255 area 0
 network 192.168.3.0 0.0.0.255 area 0
 network 192.168.7.0 0.0.0.255 area 0
```

Show IP OSPF Interface

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```

GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.3.1/24, Area 0
  Process ID 30, Router ID 3.3.3.3, Network Type BROADCAST,
Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 192.168.3.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Internet address is 192.168.7.1/24, Area 0
  Process ID 30, Router ID 3.3.3.3, Network Type POINT-TO-POINT,
Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Hello due in 00:00:05
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.7.2
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Internet address is 192.168.6.2/24, Area 0
  Process ID 30, Router ID 3.3.3.3, Network Type POINT-TO-POINT,
Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
  Hello due in 00:00:04
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)

```

Show IP OSPF Neighbor

Neighbor ID	Pri	State	Dead Time	Address
Interface				
192.168.7.2	0	FULL/ -	00:00:39	192.168.7.2
Serial0/1/0				
2.2.2.2	0	FULL/ -	00:00:33	192.168.6.1
Serial0/1/1				

Show IP Route

Gateway of last resort is not set

```
O   192.168.1.0/24 [110/129] via 192.168.6.1, 00:15:10,
Serial0/1/1
O   192.168.2.0/24 [110/65] via 192.168.6.1, 00:15:10,
Serial0/1/1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.3.0/24 is directly connected,
GigabitEthernet0/0/0
L   192.168.3.1/32 is directly connected,
GigabitEthernet0/0/0
O   192.168.4.0/24 [110/65] via 192.168.7.2, 00:15:10,
Serial0/1/0
O   192.168.5.0/24 [110/128] via 192.168.6.1, 00:15:10,
Serial0/1/1
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.6.0/24 is directly connected, Serial0/1/1
L   192.168.6.2/32 is directly connected, Serial0/1/1
    192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.7.0/24 is directly connected, Serial0/1/0
L   192.168.7.1/32 is directly connected, Serial0/1/0
```

● Router 4

Show Run

```
interface GigabitEthernet0/0/0
 ip address 192.168.4.1 255.255.255.0
 duplex auto
 speed auto

interface Serial0/1/0
 ip address 192.168.7.2 255.255.255.0
 clock rate 56000

router ospf 40
```

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```
log-adjacency-changes
network 192.168.4.0 0.0.0.255 area 0
network 192.168.7.0 0.0.0.255 area 0
```

Show IP OSPF Interface

```
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.4.1/24, Area 0
  Process ID 40, Router ID 192.168.7.2, Network Type BROADCAST,
Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.7.2, Interface address
192.168.4.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:03
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Internet address is 192.168.7.2/24, Area 0
  Process ID 40, Router ID 192.168.7.2, Network Type
POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40,
Retransmit 5
    Hello due in 00:00:09
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
```

Show IP OSPF Neighbor

Neighbor ID	Pri	State	Dead Time	Address
Interface				
3.3.3.3	0	FULL/ -	00:00:32	192.168.7.1
Serial0/1/0				

Show IP Route

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Gateway of last resort is not set

```
O    192.168.1.0/24 [110/193] via 192.168.7.1, 00:19:13,
Serial0/1/0
O    192.168.2.0/24 [110/129] via 192.168.7.1, 00:19:23,
Serial0/1/0
O    192.168.3.0/24 [110/65] via 192.168.7.1, 00:19:23,
Serial0/1/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected,
GigabitEthernet0/0/0
L    192.168.4.1/32 is directly connected,
GigabitEthernet0/0/0
O    192.168.5.0/24 [110/192] via 192.168.7.1, 00:19:23,
Serial0/1/0
O    192.168.6.0/24 [110/128] via 192.168.7.1, 00:19:23,
Serial0/1/0
    192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.7.0/24 is directly connected, Serial0/1/0
L    192.168.7.2/32 is directly connected, Serial0/1/0
```

- **PC1**

Ping R1

```
C:\>ping 192.168.1.1
```

Pinging 192.168.1.1 with 32 bytes of data:

```
Reply from 192.168.1.1: bytes=32 time=10ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
```

Ping statistics for 192.168.1.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

```
C:\>ping 192.168.5.1
```

Pinging 192.168.5.1 with 32 bytes of data:

```
Reply from 192.168.5.1: bytes=32 time=1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
```

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Ping statistics for 192.168.5.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

Ping R2

Ping statistics for 192.168.5.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=2ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=2ms TTL=254

Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.168.6.1

Pinging 192.168.6.1 with 32 bytes of data:

Reply from 192.168.6.1: bytes=32 time=41ms TTL=254
Reply from 192.168.6.1: bytes=32 time=1ms TTL=254
Reply from 192.168.6.1: bytes=32 time=3ms TTL=254
Reply from 192.168.6.1: bytes=32 time=3ms TTL=254

Ping statistics for 192.168.6.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 41ms, Average = 12ms

Ping R3

C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time=3ms TTL=253

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```
Reply from 192.168.6.2: bytes=32 time=7ms TTL=253
Reply from 192.168.6.2: bytes=32 time=2ms TTL=253
Reply from 192.168.6.2: bytes=32 time=4ms TTL=253
```

Ping statistics for 192.168.6.2:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 4ms
```

```
C:\>ping 192.168.3.1
```

Pinging 192.168.3.1 with 32 bytes of data:

```
Reply from 192.168.3.1: bytes=32 time=2ms TTL=253
Reply from 192.168.3.1: bytes=32 time=6ms TTL=253
Reply from 192.168.3.1: bytes=32 time=2ms TTL=253
Reply from 192.168.3.1: bytes=32 time=4ms TTL=253
```

Ping statistics for 192.168.3.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 3ms
```

```
C:\>ping 192.168.7.1
```

Pinging 192.168.7.1 with 32 bytes of data:

```
Reply from 192.168.7.1: bytes=32 time=62ms TTL=253
Reply from 192.168.7.1: bytes=32 time=12ms TTL=253
Reply from 192.168.7.1: bytes=32 time=37ms TTL=253
Reply from 192.168.7.1: bytes=32 time=2ms TTL=253
```

Ping statistics for 192.168.7.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 62ms, Average = 28ms
```

Ping R4

```
C:\>ping 192.168.7.2
```

Pinging 192.168.7.2 with 32 bytes of data:

```
Reply from 192.168.7.2: bytes=32 time=79ms TTL=252
Reply from 192.168.7.2: bytes=32 time=16ms TTL=252
Reply from 192.168.7.2: bytes=32 time=18ms TTL=252
Reply from 192.168.7.2: bytes=32 time=3ms TTL=252
```

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Ping statistics for 192.168.7.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 79ms, Average = 29ms

C:\>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:

Reply from 192.168.4.1: bytes=32 time=56ms TTL=252
Reply from 192.168.4.1: bytes=32 time=5ms TTL=252
Reply from 192.168.4.1: bytes=32 time=9ms TTL=252
Reply from 192.168.4.1: bytes=32 time=8ms TTL=252

Ping statistics for 192.168.4.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 56ms, Average = 19ms

Ping PC2

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=18ms TTL=126
Reply from 192.168.2.3: bytes=32 time=33ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 33ms, Average = 17ms

Ping PC3

C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.3: bytes=32 time=12ms TTL=125
Reply from 192.168.3.3: bytes=32 time=10ms TTL=125
Reply from 192.168.3.3: bytes=32 time=15ms TTL=125

Ping statistics for 192.168.3.3:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 15ms, Average = 12ms

Ping PC4

C:\>ping 192.168.4.3

Pinging 192.168.4.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.3: bytes=32 time=15ms TTL=124
Reply from 192.168.4.3: bytes=32 time=3ms TTL=124
Reply from 192.168.4.3: bytes=32 time=14ms TTL=124

Ping statistics for 192.168.4.3:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 15ms, Average = 10ms

• PC2

Ping R1

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=254
Reply from 192.168.1.1: bytes=32 time=1ms TTL=254
Reply from 192.168.1.1: bytes=32 time=40ms TTL=254
Reply from 192.168.1.1: bytes=32 time=5ms TTL=254

Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 40ms, Average = 11ms

C:\>ping 192.168.5.1

Pinging 192.168.5.1 with 32 bytes of data:

Reply from 192.168.5.1: bytes=32 time=1ms TTL=254
Reply from 192.168.5.1: bytes=32 time=7ms TTL=254
Reply from 192.168.5.1: bytes=32 time=3ms TTL=254
Reply from 192.168.5.1: bytes=32 time=1ms TTL=254

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```
Ping statistics for 192.168.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 3ms
```

Ping R2

```
C:\>ping 192.168.5.2
```

```
Pinging 192.168.5.2 with 32 bytes of data:
```

```
Reply from 192.168.5.2: bytes=32 time=1ms TTL=255
Reply from 192.168.5.2: bytes=32 time<1ms TTL=255
Reply from 192.168.5.2: bytes=32 time<1ms TTL=255
Reply from 192.168.5.2: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.2.1
```

```
Pinging 192.168.2.1 with 32 bytes of data:
```

```
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.6.1
```

```
Pinging 192.168.6.1 with 32 bytes of data:
```

```
Reply from 192.168.6.1: bytes=32 time=1ms TTL=255
Reply from 192.168.6.1: bytes=32 time<1ms TTL=255
Reply from 192.168.6.1: bytes=32 time<1ms TTL=255
Reply from 192.168.6.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.6.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

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Minimum = 0ms, Maximum = 1ms, Average = 0ms

Ping R3

C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time=2ms TTL=254
Reply from 192.168.6.2: bytes=32 time=1ms TTL=254
Reply from 192.168.6.2: bytes=32 time=3ms TTL=254
Reply from 192.168.6.2: bytes=32 time=2ms TTL=254

Ping statistics for 192.168.6.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=2ms TTL=254
Reply from 192.168.3.1: bytes=32 time=2ms TTL=254
Reply from 192.168.3.1: bytes=32 time=3ms TTL=254
Reply from 192.168.3.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.3.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>ping 192.168.7.1

Pinging 192.168.7.1 with 32 bytes of data:

Reply from 192.168.7.1: bytes=32 time=1ms TTL=254
Reply from 192.168.7.1: bytes=32 time=3ms TTL=254
Reply from 192.168.7.1: bytes=32 time=4ms TTL=254
Reply from 192.168.7.1: bytes=32 time=5ms TTL=254

Ping statistics for 192.168.7.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 5ms, Average = 3ms

Ping R4

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```
C:\>ping 192.168.7.2
```

```
Pinging 192.168.7.2 with 32 bytes of data:
```

```
Reply from 192.168.7.2: bytes=32 time=3ms TTL=253
Reply from 192.168.7.2: bytes=32 time=2ms TTL=253
Reply from 192.168.7.2: bytes=32 time=7ms TTL=253
Reply from 192.168.7.2: bytes=32 time=4ms TTL=253
```

```
Ping statistics for 192.168.7.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 4ms
```

```
C:\>ping 192.168.4.1
```

```
Pinging 192.168.4.1 with 32 bytes of data:
```

```
Reply from 192.168.4.1: bytes=32 time=2ms TTL=253
Reply from 192.168.4.1: bytes=32 time=2ms TTL=253
Reply from 192.168.4.1: bytes=32 time=4ms TTL=253
Reply from 192.168.4.1: bytes=32 time=10ms TTL=253
```

```
Ping statistics for 192.168.4.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 4ms
```

Ping PC1

```
C:\>ping 192.168.1.3
```

```
Pinging 192.168.1.3 with 32 bytes of data:
```

```
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=12ms TTL=126
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=5ms TTL=126
```

```
Ping statistics for 192.168.1.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 5ms
```

Ping PC3

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```
C:\>ping 192.168.3.3
```

```
Pinging 192.168.3.3 with 32 bytes of data:
```

```
Reply from 192.168.3.3: bytes=32 time=2ms TTL=126  
Reply from 192.168.3.3: bytes=32 time=12ms TTL=126  
Reply from 192.168.3.3: bytes=32 time=3ms TTL=126  
Reply from 192.168.3.3: bytes=32 time=12ms TTL=126
```

```
Ping statistics for 192.168.3.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 12ms, Average = 7ms
```

Ping PC4

```
C:\>ping 192.168.4.3
```

```
Pinging 192.168.4.3 with 32 bytes of data:
```

```
Reply from 192.168.4.3: bytes=32 time=3ms TTL=125  
Reply from 192.168.4.3: bytes=32 time=11ms TTL=125  
Reply from 192.168.4.3: bytes=32 time=14ms TTL=125  
Reply from 192.168.4.3: bytes=32 time=15ms TTL=125
```

```
Ping statistics for 192.168.4.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 3ms, Maximum = 15ms, Average = 10ms
```

• PC3

Ping R1

```
C:\>ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time=52ms TTL=253  
Reply from 192.168.1.1: bytes=32 time=2ms TTL=253  
Reply from 192.168.1.1: bytes=32 time=2ms TTL=253  
Reply from 192.168.1.1: bytes=32 time=15ms TTL=253
```

```
Ping statistics for 192.168.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 52ms, Average = 17ms
```

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```
C:\>ping 192.168.5.1
```

```
Pinging 192.168.5.1 with 32 bytes of data:
```

```
Reply from 192.168.5.1: bytes=32 time=3ms TTL=253  
Reply from 192.168.5.1: bytes=32 time=4ms TTL=253  
Reply from 192.168.5.1: bytes=32 time=11ms TTL=253  
Reply from 192.168.5.1: bytes=32 time=2ms TTL=253
```

```
Ping statistics for 192.168.5.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 11ms, Average = 5ms
```

Ping R2

```
C:\>ping 192.168.5.2
```

```
Pinging 192.168.5.2 with 32 bytes of data:
```

```
Reply from 192.168.5.2: bytes=32 time=2ms TTL=254  
Reply from 192.168.5.2: bytes=32 time=1ms TTL=254  
Reply from 192.168.5.2: bytes=32 time=1ms TTL=254  
Reply from 192.168.5.2: bytes=32 time=1ms TTL=254
```

```
Ping statistics for 192.168.5.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\>ping 192.168.2.1
```

```
Pinging 192.168.2.1 with 32 bytes of data:
```

```
Reply from 192.168.2.1: bytes=32 time=41ms TTL=254  
Reply from 192.168.2.1: bytes=32 time=2ms TTL=254  
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254  
Reply from 192.168.2.1: bytes=32 time=2ms TTL=254
```

```
Ping statistics for 192.168.2.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 1ms, Maximum = 41ms, Average = 11ms
```

```
C:\>ping 192.168.6.1
```

```
Pinging 192.168.6.1 with 32 bytes of data:
```

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```
Reply from 192.168.6.1: bytes=32 time=2ms TTL=254
Reply from 192.168.6.1: bytes=32 time=1ms TTL=254
Reply from 192.168.6.1: bytes=32 time=1ms TTL=254
Reply from 192.168.6.1: bytes=32 time=3ms TTL=254
```

Ping statistics for 192.168.6.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Ping R3

```
C:\>ping 192.168.6.2
```

Pinging 192.168.6.2 with 32 bytes of data:

```
Reply from 192.168.6.2: bytes=32 time=2ms TTL=255
Reply from 192.168.6.2: bytes=32 time<1ms TTL=255
Reply from 192.168.6.2: bytes=32 time<1ms TTL=255
Reply from 192.168.6.2: bytes=32 time<1ms TTL=255
```

Ping statistics for 192.168.6.2:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

```
C:\>ping 192.168.3.1
```

Pinging 192.168.3.1 with 32 bytes of data:

```
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
```

Ping statistics for 192.168.3.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.7.1
```

Pinging 192.168.7.1 with 32 bytes of data:

```
Reply from 192.168.7.1: bytes=32 time=1ms TTL=255
Reply from 192.168.7.1: bytes=32 time<1ms TTL=255
```

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```
Reply from 192.168.7.1: bytes=32 time<1ms TTL=255
Reply from 192.168.7.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.7.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping R4

```
C:\>ping 192.168.7.2
```

```
Pinging 192.168.7.2 with 32 bytes of data:
```

```
Reply from 192.168.7.2: bytes=32 time=16ms TTL=254
Reply from 192.168.7.2: bytes=32 time=1ms TTL=254
Reply from 192.168.7.2: bytes=32 time=10ms TTL=254
Reply from 192.168.7.2: bytes=32 time=3ms TTL=254
```

```
Ping statistics for 192.168.7.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 7ms
```

```
C:\>ping 192.168.4.1
```

```
Pinging 192.168.4.1 with 32 bytes of data:
```

```
Reply from 192.168.4.1: bytes=32 time=11ms TTL=254
Reply from 192.168.4.1: bytes=32 time=1ms TTL=254
Reply from 192.168.4.1: bytes=32 time=1ms TTL=254
Reply from 192.168.4.1: bytes=32 time=1ms TTL=254
```

```
Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms
```

Ping PC1

```
C:\>ping 192.168.1.3
```

```
Pinging 192.168.1.3 with 32 bytes of data:
```

```
Reply from 192.168.1.3: bytes=32 time=52ms TTL=125
Reply from 192.168.1.3: bytes=32 time=19ms TTL=125
Reply from 192.168.1.3: bytes=32 time=7ms TTL=125
```

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Reply from 192.168.1.3: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.1.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 7ms, Maximum = 52ms, Average = 22ms

Ping PC2

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=7ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.2.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 12ms, Average = 5ms

Ping PC4

C:\>ping 192.168.4.3

Pinging 192.168.4.3 with 32 bytes of data:

Reply from 192.168.4.3: bytes=32 time=2ms TTL=126
Reply from 192.168.4.3: bytes=32 time=12ms TTL=126
Reply from 192.168.4.3: bytes=32 time=11ms TTL=126
Reply from 192.168.4.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 12ms, Average = 6ms

● PC4

Ping R1

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

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```
Reply from 192.168.1.1: bytes=32 time=47ms TTL=252
Reply from 192.168.1.1: bytes=32 time=18ms TTL=252
Reply from 192.168.1.1: bytes=32 time=3ms TTL=252
Reply from 192.168.1.1: bytes=32 time=3ms TTL=252
```

Ping statistics for 192.168.1.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 47ms, Average = 17ms
```

```
C:\>ping 192.168.5.1
```

Pinging 192.168.5.1 with 32 bytes of data:

```
Reply from 192.168.5.1: bytes=32 time=52ms TTL=252
Reply from 192.168.5.1: bytes=32 time=11ms TTL=252
Reply from 192.168.5.1: bytes=32 time=3ms TTL=252
Reply from 192.168.5.1: bytes=32 time=23ms TTL=252
```

Ping statistics for 192.168.5.1:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 52ms, Average = 22ms
```

Ping R2

```
C:\>ping 192.168.5.2
```

Pinging 192.168.5.2 with 32 bytes of data:

```
Reply from 192.168.5.2: bytes=32 time=3ms TTL=253
Reply from 192.168.5.2: bytes=32 time=6ms TTL=253
Reply from 192.168.5.2: bytes=32 time=2ms TTL=253
Reply from 192.168.5.2: bytes=32 time=6ms TTL=253
```

Ping statistics for 192.168.5.2:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 4ms
```

```
C:\>ping 192.168.2.1
```

Pinging 192.168.2.1 with 32 bytes of data:

```
Reply from 192.168.2.1: bytes=32 time=4ms TTL=253
Reply from 192.168.2.1: bytes=32 time=2ms TTL=253
Reply from 192.168.2.1: bytes=32 time=2ms TTL=253
```

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Reply from 192.168.2.1: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.2.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\>ping 192.168.6.1

Pinging 192.168.6.1 with 32 bytes of data:

Reply from 192.168.6.1: bytes=32 time=3ms TTL=253
Reply from 192.168.6.1: bytes=32 time=4ms TTL=253
Reply from 192.168.6.1: bytes=32 time=15ms TTL=253
Reply from 192.168.6.1: bytes=32 time=6ms TTL=253

Ping statistics for 192.168.6.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 15ms, Average = 7ms

Ping R3

C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time=2ms TTL=254
Reply from 192.168.6.2: bytes=32 time=11ms TTL=254
Reply from 192.168.6.2: bytes=32 time=1ms TTL=254
Reply from 192.168.6.2: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.6.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time=2ms TTL=254
Reply from 192.168.3.1: bytes=32 time=3ms TTL=254
Reply from 192.168.3.1: bytes=32 time=18ms TTL=254
Reply from 192.168.3.1: bytes=32 time=4ms TTL=254

Ping statistics for 192.168.3.1:

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Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 18ms, Average = 6ms

C:\>ping 192.168.7.1

Pinging 192.168.7.1 with 32 bytes of data:

Reply from 192.168.7.1: bytes=32 time=2ms TTL=254
Reply from 192.168.7.1: bytes=32 time=1ms TTL=254
Reply from 192.168.7.1: bytes=32 time=2ms TTL=254
Reply from 192.168.7.1: bytes=32 time=2ms TTL=254

Ping statistics for 192.168.7.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

Ping R4

C:\>ping 192.168.7.2

Pinging 192.168.7.2 with 32 bytes of data:

Reply from 192.168.7.2: bytes=32 time=2ms TTL=255
Reply from 192.168.7.2: bytes=32 time=1ms TTL=255
Reply from 192.168.7.2: bytes=32 time<1ms TTL=255
Reply from 192.168.7.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.7.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms

Pinging 192.168.4.1 with 32 bytes of data:

Reply from 192.168.4.1: bytes=32 time=1ms TTL=255
Reply from 192.168.4.1: bytes=32 time<1ms TTL=255
Reply from 192.168.4.1: bytes=32 time<1ms TTL=255
Reply from 192.168.4.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.4.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

Ping PC1

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```
C:\>ping 192.168.1.3
```

```
Pinging 192.168.1.3 with 32 bytes of data:
```

```
Reply from 192.168.1.3: bytes=32 time=91ms TTL=124
Reply from 192.168.1.3: bytes=32 time=11ms TTL=124
Reply from 192.168.1.3: bytes=32 time=15ms TTL=124
Reply from 192.168.1.3: bytes=32 time=17ms TTL=124
```

```
Ping statistics for 192.168.1.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 91ms, Average = 33ms
```

Ping PC2

```
C:\>ping 192.168.2.3
```

```
Pinging 192.168.2.3 with 32 bytes of data:
```

```
Reply from 192.168.2.3: bytes=32 time=66ms TTL=125
Reply from 192.168.2.3: bytes=32 time=3ms TTL=125
Reply from 192.168.2.3: bytes=32 time=43ms TTL=125
Reply from 192.168.2.3: bytes=32 time=2ms TTL=125
```

```
Ping statistics for 192.168.2.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 66ms, Average = 28ms
```

Ping PC3

```
C:\>ping 192.168.3.3
```

```
Pinging 192.168.3.3 with 32 bytes of data:
```

```
Reply from 192.168.3.3: bytes=32 time=1ms TTL=126
Reply from 192.168.3.3: bytes=32 time=52ms TTL=126
Reply from 192.168.3.3: bytes=32 time=13ms TTL=126
Reply from 192.168.3.3: bytes=32 time=13ms TTL=126
```

```
Ping statistics for 192.168.3.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 52ms, Average = 19ms
```

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Problems

One issue I had was that I originally only made four networks, which was not enough to account for each of the interface connections to other devices. Without enough networks, I was not able to get OSPF working. I changed the IP addresses on specific interfaces and created three more networks to fix this issue.

Another problem I had was that I did not subnet the networks correctly. Some networks were /30 and others were /29. I was confused on the number of hosts available when using these subnet masks. Initially, I tried to have the networks only have a limited number of hosts available, but when I was done I could not ping any device from another device – not even router to router or router to PC on the same network. To solve this, I redid the subnets on all the devices to /24 to make certain that there were enough available hosts for all the devices, interfaces in each network, and to make it less confusing for me while keeping the rest of the configurations in my lab the same. After this, I was able to ping every device to each of the other devices.

Conclusion

Setting up a single-area OSPF using seven networks, four routers, and four computers was a good refresher for me to recall how to build a network, connect multiple devices, diagnose any problems, and resolve those issues. This lab reminded me what kind of mistakes I am susceptible to and how I can fix those problems, which ensures I will be more aware of similar mistakes. They say people learn by doing and that people learn history to learn from other's past mistakes. This also applies to CISCO labs like this one, which taught me how to set up a working single-area OSPF configuration that I can use as reference for future labs and to not repeat the issues I created in this set up. Additionally, this was the first lab where I started from scratch, building my own topology, determining what subnet and IP addresses to use, and figuring out how many network I need, while the previously I mostly worked with pre-set topologies, IP addresses, and number of networks.